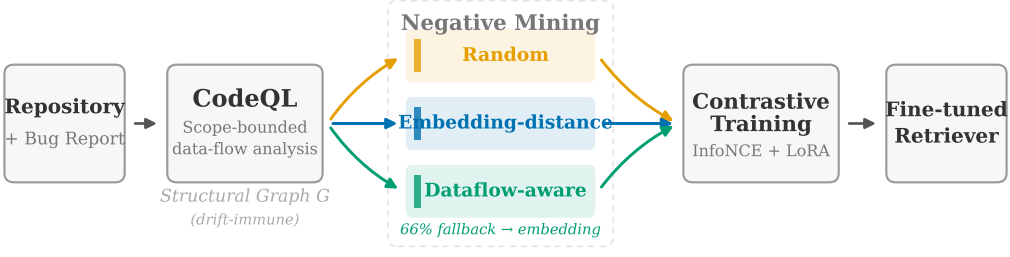
Repository
+ Bug Report
CodeQL
Scope-bounded data-flow analysis
Structural Graph G
(drift-immune)
Negative Mining
Random
Embedding-distance
Dataflow-aware
66% fallback → embedding
Contrastive Training
InfoNCE + LoRA
Fine-tuned Retriever
Static mining — computed once, no re-mining during training